R16

Code No: 135CX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to CE, EEE, CSE, EIE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Explain the frequency translation concept with examples. [15]

2. Describe the need for modulation, in detail with examples. [15]

3.a) Define PWM and explain the modulation process with neat diagram.
b) Explain AM with spectrum. Show that non-linear device can be used for generating AM signal. What are its limitations? [7+8]

4.a) An AM signal is represented by $v(t) = 0.1(1+0.1\cos(2512t)+0.5\cos(6280t))\sin(10^7+45^0)$ volts. What information can you get from this? Plot the amplitude spectrum of the signal.
b) Explain the indirect method of generation of FM wave. [7+8]

5.a) Draw the general block diagram of cordless telephone and explain.
b) Describe token ring LAN. [8+7]

6.a) What are the components of a facsimile system and explain each one in detail.
b) Discuss about paging systems in detail. [8+7]

7.a) Discuss the importance of GPs with respect to satellite communication.
b) Explain different fiber optic cables used in optical communications. [7+8]

8.a) Explain about CDMA.
b) Discuss about Network and Switching Subsystem. [8+7]

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